Three Reasons AR/VR Telesurgery Should Lead Your Surgical Strategy

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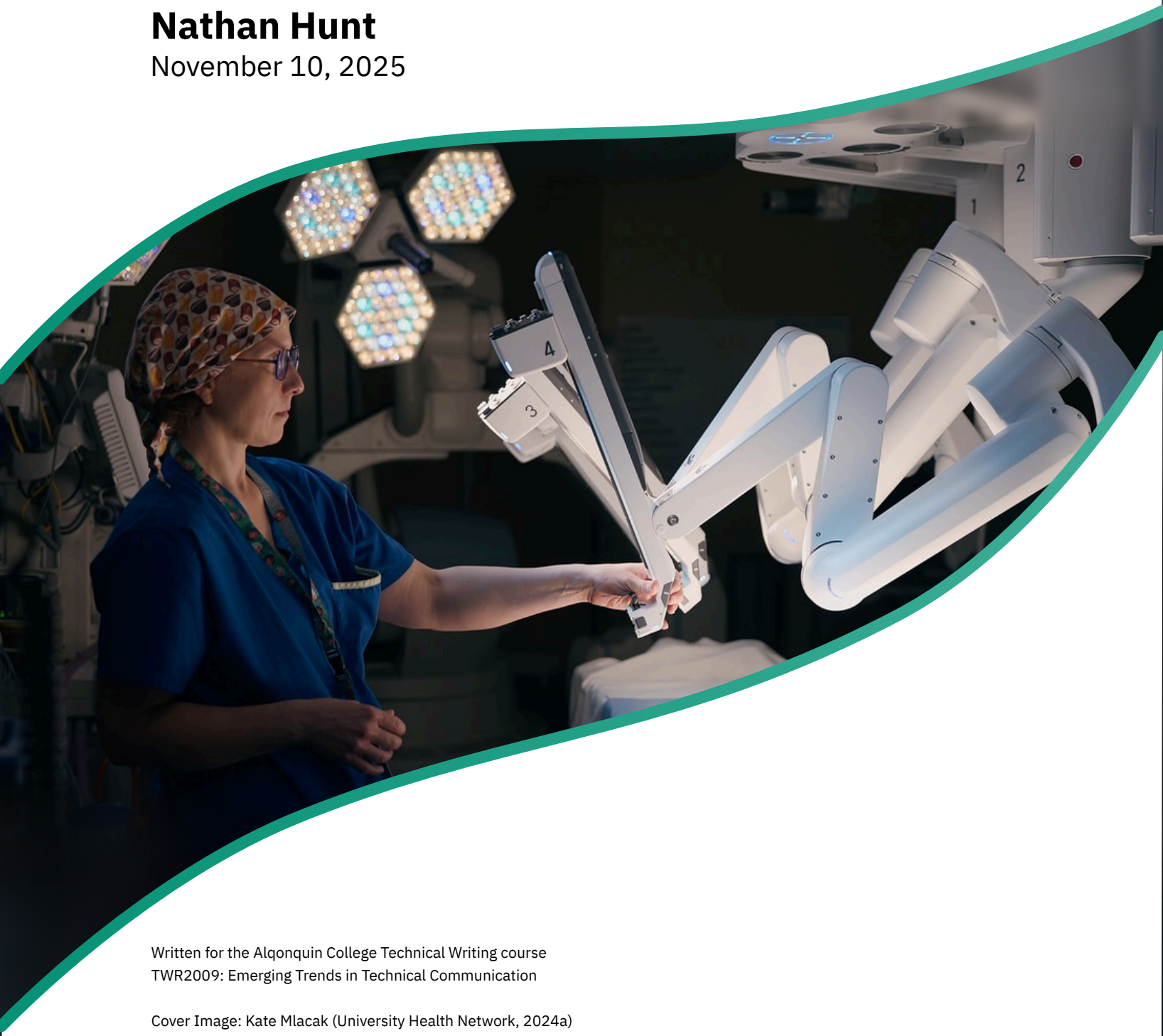

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Cover Image: Kate Mlacak (University Health Network, 2024a)

Barriers to Surgical Innovation

This paper is primarily for doctors, directors, and administrators of Canadian hospitals or regional health-care networks. Your organization may be exploring ways to improve your surgical services that are innovative, equitable and cost-effective. In this paper we will explore the use of Virtual Reality (VR) and Augmented Reality (AR) in robotic-assisted surgery, and the benefits it brings for patients, doctors, and institutions.

Today's hospitals and health networks face many hurdles to providing the level of surgical care that patients need. Canada's geography and low population density leave certain communities underserved and with limited access to surgical care. When organizations are under-funded and hallway medicine has become routine, it can be difficult to push for major investments in novel technologies. However, Canada's reputation as a global health-care leader is at risk, as we continue to fall behind in global health metrics (Longhurst, 2025). The challenges organizations face also represent a strategic opportunity: your institution can lead the way by developing AR robotic telesurgery as the future of surgery in Canada.

What is telerobotic surgery?

Telerobotic surgery is an extension of Robotic Assisted Surgery (RAS). In RAS, a set of robotic manipulators perform the operation. These systems do not act autonomously; rather, they work as an extension of the tools controlled by the surgeon (Morris, 2005). A key feature of RAS is the physical separation of the surgical controller and the robotic manipulator. With telerobotic surgery, the surgeon can operate on a patient hundreds of kilometres away.

Key terms

Augmented Reality (AR): A computer-rendered environment projected into the real world with a virtual reality headset.

Haptic feedback (also, haptics): Using physical resistance or vibration to engage a user's sense of touch. Haptics give physical feedback from a virtual world (Vincent, 2018).

Robotic manipulators: Also referred to a "slave equipment" or "slave robotic manipulators." The set of robotic surgical tools and implements that operate on a patient (Morris, 2005).

Robotic Minimally Invasive Surgery (RMIS): The robotic-assisted version of Minimally Invasive Surgery (MIS). The operation is done with tools and cameras inserted through small incisions in the skin instead of open surgery.

Surgical controller: Also called a "master console" or "master surgical controller." The set of controls and displays used by the surgeon to control the instruments (Morris, 2005).

Telesurgery (also, telerobotic surgery): Surgery, pre- and post-operative work,done by a surgeon physically separated from the patient (Sherif et al., 2023).

Virtual Reality (VR): A computer-generated environment that immerses a viewer, usually with a stereoscopic headset.

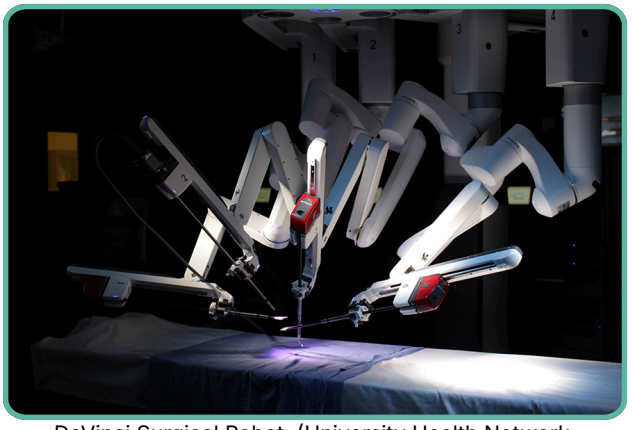
History of robotic surgery

1980s	Early robotic surgical systems developed (Sherif et al., 2023).
1997	First robotic cholecystectomy performed (Morris, 2005).
2001	First remote telesurgery takes place between New York and Strasbourg (Choi et al., 2018).
2003	Telesurgical service established between St. Joseph's Hospital, Hamilton, and North Bay General Hospital (Anvari et al., 2005).
2014	Surgeons in four cities use AR to view and interact with a surgical field at the same time (Choi et al., 2018).
2018	Fundamental VR creates surgical training tools with consumer VR headset (Vincent, 2018).
2024	University Health Network expands RAS capacity with two additional robots (Danko, 2024)

The Three Reasons to Invest in VR/AR Telesurgery

1. Robotic surgery improves patient outcomes

Robotic-assisted surgery and Robotic Minimally Invasive Surgery (RMIS) is used widely for urological, gynecological, cardiac, and general surgeries. There are at least 39 sites across Canada that perform RAS procedures. Many laparoscopic surgeries—done with endoscopic cameras and instruments inserted through a small incision—are being converted to RMIS (Wong et al., 2025). Some medical experts believe that nearly all surgical procedures will be one day be robotic assisted (Morris, 2005).



DaVinci Surgical Robot. (University Health Network, 2024b)

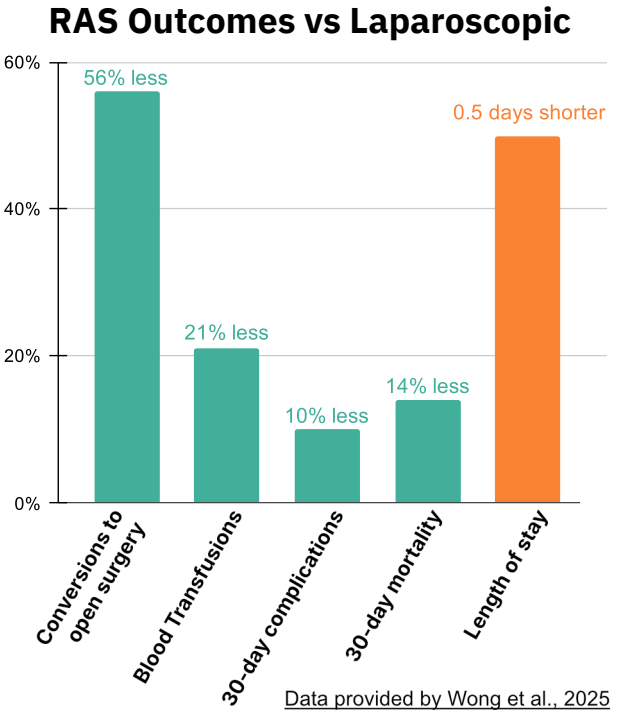
RMIS benefits surgeons by allowing them to operate from a comfortable position, reducing fatigue. Robotic tools give surgeons an improved range of motion, greater precision, and effectively eliminate the vibrations of human hands (Morris, 2005). All these factors make surgeons more effective at providing top-quality care.

> **RAS can result in fewer blood transfusions, lower rates of 30-day complications, and fewer 30-day readmissions.**
> (Wong et al., 2025)

Studies have found that robotic assisted surgeries have better patient outcomes than traditional open or laparoscopic procedures, without a significant increase in risk. A meta-study on the impact of RAS procedures in Canada found that:

- RAS increased minimally invasive surgery rates from 47% to 85% for colorectal cancer cases.
- RAS had 9% fewer 30-day readmissions when compared to laparoscopic surgery.
- Robotic lung resection procedures had lower rates of respiratory failure.

Other benefits of RAS for patients include quicker recovery times, lower readmission rates, reduced wound complications, and less pain. Hospitals can discharge patients earlier and allow them to recover at home, reducing the patient's stress, anxiety and financial burden (Wong et al., 2025).



Data provided by Wong et al., 2025

2. Telerobotics makes surgery more accessible

A significant number of Canadians face barriers to accessing health services, brought about by geographic limitations, socio-economic factors, and overburdened health facilities. In 2024 almost half of Ontario's non-profit hospitals faced operating budget deficits. In 2022, Canada ranked lower than 30 of the 38 OECD countries in number of hospital beds per-capita (Longhurst, 2025).

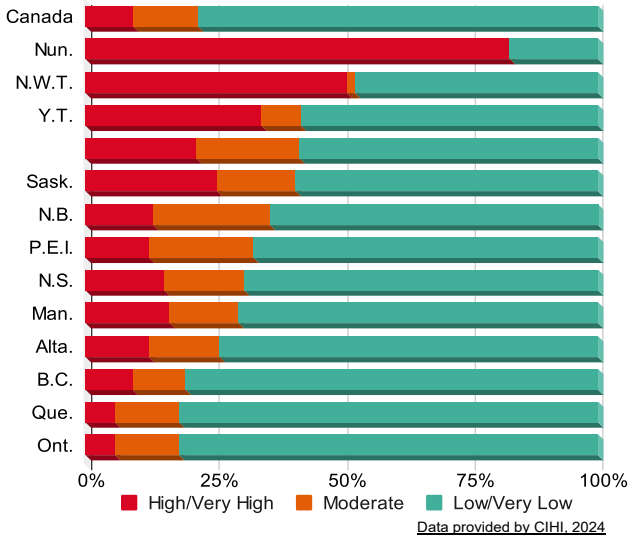
Telerobotic surgery can relieve an over-capacity hospital by improving patient flow. A 2025 paper found that the shorter hospital stays and faster recovery times of RAS lead to an overall cost savings and improved efficiency. Telesurgery can also eliminate the risks and costs of transporting ill or injured patients from one facility to another (Liu et al., 2024; Wong et al., 2025).

Travel Burden:
A complex metric factoring in distance travelled to and from hospital, options for transportation, time and costs associated with travel, and opportunity costs (Canadian Institute for Health Information [CIHI], 2024).

Canadians living in northern and rural communities are less likely to have a family doctor and often travel long distances to access health services (Canadian Medical Association, 2014). Studies show that travel burden is disproportionately high in the territories: 83% of the population of Nunavut face a high or very high travel burden, and 19% of Ontarians have a moderate or higher travel burden (CIHI, 2024).

VR/AR telerobotic surgery has the potential to break down travel barriers by establishing remote surgical suites in underserved areas. By building a telerobotic surgery network, you can expand your organization's service area at a comparatively low cost. Telesurgery also gives your patients access to a larger pool of specialists.

Travel Burden Across Canada (% of pop.)



Data provided by CIHI, 2024

By building a telerobotic surgery network you can expand your organization's service area at a comparatively low cost.

There are still challenges to establishing a VR/AR telesurgery network in Canada. Estimates put the initial cost of a telesurgery system at $1,200,000, with operating costs around $100,000 per year (Morris, 2005). Establishing surgical facilities in remote locations could cost much more.

To manage the capital costs, you may be able to leverage partnerships with larger healthcare networks or other institutions to pool resources (Singal, 2019). Some robotics companies offer alternative funding models that factor the installation costs into your operating budget, eliminating the need to raise funding up-front (Wong et al., 2025).

3. Telesurgery technology is advancing rapidly

With the widespread adoption of robotic-assisted surgery, telerobotics technologies continue to improve. The next innovation is with applying AR technologies to surgical controllers. VR/AR headsets are getting lighter and more ergonomic, making them comfortable enough to wear for an entire operation. Haptic feedback using resistance and vibration gives direct and immediate feedback, immersing the surgeon in a virtual world (Vincent, 2018).

Life-like anatomic rendering. (Fundamental XR, n.d. – a)

Telesurgery requires stable and fast network connections between the surgical controller and robotic manipulators; a lag time of more than 200 milliseconds can degrade the surgeon's performance (Sherif et al., 2023). This made setting up telesurgery sites in remote areas almost impossible, but today's satellite internet services can easily achieve the connection speeds required.

Modern graphics engines can render life-like scenes in real-time (Fundamental XR, n.d. – b). A future surgeon may operate from a workstation wearing a headset that merges digital and physical worlds, using virtual tools as if they held the scalpel in their hands.

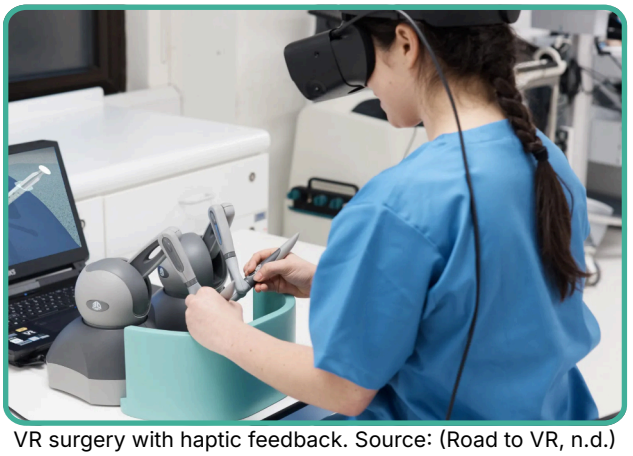
Recent advancements in robotic manipulators and surgical tools include:

- Machine learning, neural networks, and computer vision.
- Robotic armatures with high degrees-of-freedom that mimic the human wrist.
- Inverse kinematic systems that keep surgical tools within safe limits.
- VR digital-twin systems for data analysis and post-operative review.
- High-resolution video feeds from endoscopic cameras (Cai et al., 2023).

Someday, a surgeon in Vancouver may be able to provide trauma care for a patient thousands of kilometres away.

By becoming a leader in VR/AR telerobotic surgery, you can develop Canada's global reputation for top-tier surgical care. Today's health-care professionals consider RAS an essential tool, and RMIS is a standard part of a surgeon's education (Wong et al., 2025). Building telesurgery capacity can attract global talent to your organization.

Canadian surgeons could provide surgical care to resource-limited areas around the globe. Someday, a surgeon in Vancouver may be able to provide trauma care for a patient thousands of kilometres away using portable telesurgery suites (Sherif et al., 2023).



VR surgery with haptic feedback. Source: (Road to VR, n.d.)

Building an Innovative Surgical System

The widespread adoption of AR telesurgery still faces major hurdles. Between cash-strapped health agencies and Canada's geographic scale, a future in which high-tech robotic surgery is standard still seems far off. Performing telerobotic surgery across provincial and international borders also raises questions of licensing and accreditation (Choi et al., 2018). Telerobotic systems can be costly, and inter- provincial and institutional partnerships are no easy path to navigate.

Rural Yukon Health-care Facility. (Kawaja, 2024)

Building this robotic future will take a significant investment in technology, training, and talent. However, telerobotics surgery systems and augmented reality equipment are becoming more affordable and can lead to cost-savings when operational efficiencies and patient outcomes are factored in (Wong et al., 2025).

If you're ready to explore how your institution can become a leader in AR telesurgery, consider starting a pilot program, or a partnership with another health-care organization. Let's imagine a future where all Canadians have access to the best surgical health system in the world.

Three Reasons AR/VR Telesurgery Should Lead Your Surgical Strategy

1 Improved Outcomes

Studies find that robotic assisted surgeries have better patient outcomes than open or laparoscopic procedures, with no significant increase in risk.

2 Greater Accessibility

22% of Canadians face barriers accessing health care. Telesurgery can expand your organization's reach and lets patients access more specialists.

3 Smarter Technology

VR/AR Telesurgery is innovating with new tools and technologies. Your institution could become a global leader in an emerging field.

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Robotic Surgery by Berkah Icon from Noun Project (CC BY 3.0)

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